Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech
Code : 104602

Subject: Digital Communication

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) The sampling theorem is essential to avoid which problem?
(i) Crosstalk (iii) Attenuation
(ii) Quantization error (iv) Aliasing

(b) In Time Division Multiplexing (TDM), multiple signals share:
(i) Power slots (iii) Time slots
(ii) Code slots (iv) Frequency slots

(c) Which modulation technique is most commonly used in CDMA?
(i) QAM (iii) ASK
(ii) PSK (iv) FSK

(d) In Quadrature Amplitude Modulation (QAM), the information is carried by:
(i) Frequency and phase (iii) Amplitude only
(ii) Amplitude and phase (iv) Phase only

(e) What does the signal-to-quantization noise ratio (SQNR) measure?
(i) Bandwidth efficiency (iii) Distortion in quantized signals
(ii) Power efficiency (iv) Signal attenuation

(f) Which line coding technique uses zero crossing for binary '1' and '0'?
(i) AMI (iii) RZ
(ii) Manchester coding (iv) NRZ

(g) Which of the following is true for Adaptive Delta Modulation (ADM)?
(i) It uses fixed bandwidth (iii) It reduces slope overload
(ii) It increases quantization noise (iv) It has fixed step size

(h) A matched filter maximizes which of the following?
(i) Transmission speed (iii) Noise
(ii) Signal-to-noise ratio (iv) Signal power

(i) Inter-symbol interference (ISI) primarily occurs due to:
(i) High bit rate (iii) Over-sampling
(ii) Low signal-to-noise ratio (iv) Improper filtering

(j) Amplitude Shift Keying (ASK) modulates which signal characteristic?
(i) Pulse width (iii) Phase
(ii) Frequency (iv) Amplitude

Q.2 (a) State and prove the sampling theorem. Why is it important in digital communication systems? **[7]**

(b) Explain the working principle of Differential Phase Shift Keying (DPSK) and how it differs from standard PSK. **[7]**

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Q.3	(a)	A signal is encoded using 8-bit PCM. Calculate the data rate if the signal is sampled at 8 kHz. What will be the data rate for 16-bit PCM at the same sampling rate?	[7]
	(b)	Derive the expression for the probability of error in a matched filter receiver.	[7]
Q.4	(a)	Discuss the concept of a baseband signal receiver and the factors affecting the probability of error in digital communication systems.	[7]
	(b)	Explain the process of quantization in a digital communication system. Differentiate between uniform and non-uniform quantization.	[7]
Q.5	(a)	A QPSK system transmits at a data rate of 2 Mbps. If the noise power spectral density is 0.5 μW/Hz and the signal power is 1 mW, calculate the bit error rate (BER) of the system.	[7]
	(b)	Discuss the principles of Code Division Multiple Access (CDMA) and how it differs from TDM and FDM.	[7]
Q.6	(a)	Derive the Nyquist criterion for zero ISI. Explain how this criterion ensures distortion-free transmission.	[7]
	(b)	Describe the block diagram of a digital communication system and explain the function of each block.	[7]
Q.7	(a)	What is aliasing? Explain the causes of aliasing and how it can be avoided in digital communication systems.	[7]
	(b)	Discuss the various types of line coding techniques used in digital communication. Compare unipolar, polar, and bipolar schemes.	[7]
Q.8	(a)	Explain Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Phase Shift Keying (PSK) with waveform examples.	[7]
	(b)	What is Minimum Shift Keying (MSK)? Explain its advantages over FSK and other digital modulation techniques.	[7]

Q.9 Write short notes on ***any two*** of the following: **[7 x 2 = 14]**

(a) Orthogonal Frequency Division Multiple Access (OFDMA).
(b) Matched filter.
(c) Coherent reception & Non-Coherent reception.
(d) Advantages of Digital communication system over Analog communication systems.

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